

# P3 - Paneling, Arrays and Attractor Façade

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

**Approach:** Parametric

Revision: 2026-09-23. Instructor review draft. Grading allocation approved; dates remain proposed for review before publication.

3.1: module, authored material, ground plane and three-point lighting. 3.2: arrays with library or student-authored materials. 3.3: attractor-driven panel façade, shown in elevation and isometric.

## 3.1 - Descriptive geometry module - material and three-point lighting

**Introduced:** Week 4 - Sep 15 / **Due:** Week 9 - Oct 20, 11:59 PM Eastern Time / **Weight:** 10%

### Description

Build a panel from a real building, using point positions you write out and faces in order. Show three parameter settings on one 17 x 11 inch sheet, with one material and the same three-point lighting on a ground plane. Include the point list, face list and parameter table. You will reuse this module on Sheets 1-2 of 3.2, and in 3.3 it becomes the element the attractors change.

### Rules & Constraints

- Choose a simple panel from a built precedent that tiles edge to edge in a 2 x 2 grid. Credit the building, architect and source.
- Define the parameters first and explain each one. Beyond overall size, include one control that changes the shape, such as an opening (aperture), a fold depth or a corner offset.
- Build an editable, closed mesh from lists of points and faces that you write out yourself. Calculate the point positions from the parameters.
- For a panel with 24 or more points, you may use a Boolean if at least 16 points that you write out yourself generate the geometry. Check this approach with your professor.
- State the ranges in which the module stays closed and tiles. Show low, middle and high values of one shape parameter.
- Use one 17 x 11 inch sheet showing three rendered versions of the panel with different parameter values. Use one material you make and the same lighting for all three: three-point lighting (key, fill and rim/back), a neutral ground plane and no entourage.
- Beside the renders, show the point list, the face list (which points each face uses) and a table of the parameter values for each version. You don't need a separate explanation of face order or a lighting diagram.
- Watch the [three-point lighting tutorial](#), also listed in Resources. Three-point lighting is about three roles the lights play; it doesn't require Blender's Point-type lights.

### Parameters

- Width, depth and height/thickness in stated model units, with a local origin and X/Y/Z axes.
- At least one control that changes the shape, beyond overall size: an opening (aperture), a fold depth, a corner offset or something similar. Name the control and list its values for the three versions.
- The valid range of every control, and how the controls depend on each other. State the limits within which the faces stay valid, the solid stays closed and neighboring panels still meet.

### Required Studies

- **Precedent and construction lists.** Credit the building, architect and image source. Show the point list with each point's number (index) and coordinates, and the face list that refers to those numbers. If you use the permitted Boolean route, show which geometry comes from at least 16 points you wrote out, and review it with your professor.

- **Three parameter settings.** Render low, middle and high values of one shape parameter from the same graph. Include a table of the values for all three. Check in your working file that the panel is closed and tiles edge to edge in a 2 x 2 grid; you don't need extra construction or tiling diagrams on the sheet.
- **One material and lighting setup.** Use the same material and lighting in all three renders. The key light shows the form, the fill controls how dark the shadows are, and the rim/back light separates the edges from the background. Keep the module on a neutral ground plane. Follow the [three-point lighting tutorial](#). No lighting diagram is needed.

## Workflow

1. Choose a panel from a real building with a repeatable outline, and identify its points, edges and faces.
2. Define the inputs, calculate the vertices and list the faces in order. Check any Boolean use with your professor under the 24-point / 16-point rule above.
3. Test closure, normals and tiling at low, middle and high settings. Fix gaps, flipped faces and intersections.
4. Make one material for the module and balance the key, fill and rim lights in a scene with a ground plane. Render all three versions the same way, then lay out the sheet with the point list, face list and parameter table.

## Deliverables

- **One 17 x 11 inch landscape module sheet:** Lastname\_Firstname\_P3a.pdf. Include the credited precedent, three rendered panel variations, a point list, a face list referencing those points and a parameter table for the three settings. Use one module material and one lighting strategy throughout.
- **Editable graph:** .blend, or an already approved .gh/.3dm route. Include the material and lighting scene used for the renders, and pack any files it uses. Keep native node names visible and explain your groups with Frames.
- Submit your files in Lastname\_Firstname\_P3a/. Save the chosen parameter settings and panel origin so you can reuse exactly the same module in P3b. No print is required for this assignment.

## Evaluation

- **Computational understanding - 30%:** explain how the parameters, point list and face list build the panel, and how a shape input changes it.
- **Visual communication - 30%:** the three renders, material, lighting, lists and parameter table reveal the geometry clearly and follow the Graphic Standards Manual.
- **Technical execution - 25%:** the panel is valid, tiles in the working file and follows the point/Boolean rule; the editable scene reproduces the submitted views.
- **Exploration - 15%:** three meaningful parameter settings test the shape control and support the selected module for later assignments.

## Before You Submit

- [ ] Real building credited; panel built from calculated points and faces in order.
- [ ] One shape control beyond overall size, tested at low, middle and high values.
- [ ] Closure, normals and edge-to-edge 2 x 2 tiling checked in the working file.
- [ ] One sheet shows three rendered variations, the point list, face list and parameter table.
- [ ] Any Boolean use has been reviewed with the professor - only for a panel with 24 or more points, with at least 16 written-out points generating the geometry.
- [ ] One module material you made and one key/fill/rim lighting setup across all versions; neutral ground plane; no entourage. No lighting diagram or material-settings table is needed.

## 3.2 - Arrays

**Introduced:** Week 6 - Sep 29 / **Due:** Week 10 - Oct 27, 11:59 PM Eastern Time / **Weight:** 10%

### Description

Build basic arrays of your panel, four façade patterns made with rules, and a volume of cubes shaped by math. Present the three sheets using your Graphic Standards Manual.

### Rules & Constraints

- Submit exactly three 17 x 11 inch landscape sheets together as one assignment.
- Sheet 1 - four diagrams: your 3.1 module alone, then in a 1D, a 2D and a 3D array. Use the same module throughout, and label the counts, spacing and axes.
- Sheet 2 - four façade-pattern diagrams: apply if/Switch rules to a 2D array of your module to make four different patterns. Each diagram names its rule and what it changes.
- Sheet 3 - a volume of cubes: make a 3D array of cubes and show it in isometric. Change the height of the cubes with a math function of both X and Y, such as sine or cosine, and explore the form it creates. Optionally, use Switch rules to make voids.
- Reuse the 3.1 panel on Sheets 1-2. On Sheet 2 you may change or remove selected copies to make a pattern. Use cubes, not the panel, on Sheet 3.
- Keep the diagrams simple, with the same view and material wherever studies are compared. Keep the lighting consistent; the four façade patterns don't need separate polished renders. Choose a suitable view for each array, and use isometric for the cube volume.
- Build the repetition from basic nodes, starting with a count small enough to check by eye; there is no minimum number of copies.
- Make every array fully parametric: expose row, column and layer counts and spacing in each used direction. Changing those inputs must rebuild the array and resize the panel or cube to fit its new cell without manual remodeling or repositioning. Preserve the intended gaps or edge contact.
- Use materials from the library the instructors provide, or make your own. Note which materials you used and any changes; pack any textures. Materials must keep the spacing, depth and repetition of the array readable. The library will be posted before this assignment is due; making your own is equally fine.

### Parameters

- Whole-number counts  $N_x$ ,  $N_y$ ,  $N_z$ ; positive spacings  $dx$ ,  $dy$ ,  $dz$ , measured between module origins; a base point and array axes. Link module width, depth and any relevant height to the cell dimensions so the geometry adapts when spacing changes. Use one module orientation on Sheet 1.
- For each façade rule: what happens when it is true and when it is false, and its controls - for example, which rows, a rotation, a move, or keep/remove.
- Cube size, number of layers, the height function and its final settings, including amplitude, frequency and phase if used. Describe any optional rule for voids.
- Where each material comes from and any changes you made. Library or your own materials are both fine; pack textures.

### Required Studies

- **Sheet 1 - Module and repetition.** Show the module, a row, a grid and a 3D array. Label the counts, spacing, axes and totals:  $N_x$ ,  $N_x \times N_y$  and  $N_x \times N_y \times N_z$ . Make the layers of the 3D array visible in the diagram.
- **Sheet 2 - Four façade patterns.** Make four distinct patterns with if/Switch rules, such as alternating rows or columns, a checkerboard or diagonal bands. Rotate, move, scale or remove the selected copies. Label each rule and what happens when it is true or false.
- **Sheet 3 - Math shapes a cube volume.** Make a 3D array of cubes whose layers rise and fall like a wave, using the Sine curve scene in the Arrays Part 2 example file as a reference. One possible formula is  $f(x, y) = A \sin(f_x x + p_x) + B \cos(f_y y + p_y)$ , with each layer at  $z = k \cdot dz + f(x, y)$ , where  $k$  is the layer number and angles are in radians. Keep several layers. Explore the settings and choose one result for the isometric. Voids made with Switch are optional.

## Workflow

1. Bring in your 3.1 module. Build a row, then a 2D and a 3D array from basic nodes, and check their counts.
2. Test at least two count and spacing settings for each array. The module or cube must adapt to its cell dimensions, and copies must keep their intended fit or gap without manually editing their geometry.
3. Make four façade patterns from the 2D array. Check that each rule changes the copies you intended.
4. Build a separate array of cubes in layers. Explore functions of X and Y, optionally add voids, and save the settings of the result you choose.
5. Check for copies that overlap or sit in the same place by mistake. Lay out the three sheets with readable views, a caption for each rule, and your materials.

## Deliverables

- **Exactly three 17 x 11 inch landscape sheets in one PDF:** Lastname\_Firstname\_P3b.pdf. They are one assignment, submitted together.
- **Sheet 1 - Four diagrams.** Your module alone, then the same module in a 1D, 2D and 3D array. Label axes, counts, spacing and array totals.
- **Sheet 2 - Four façade patterns.** Four diagrams of your module in a 2D array, changed by if/Switch rules. Add a short caption to each pattern: the rule and what it does.
- **Sheet 3 - Cube volume.** One isometric view of a 3D array of cubes whose height changes with a function of X and Y. Include the formula and final settings, and describe any optional rule for voids.
- **Editable graph** in Lastname\_Firstname\_P3b/, with labeled inputs, saved settings and outputs for all three sheets. An instructor must be able to change counts and spacing and see the arrays and modules update without manual edits. Pack the files it needs and the textures, and add a note on where the materials come from. No print is required.

## Evaluation

- **Computational understanding - 30%:** explain repetition in 1D, 2D and 3D, your if/Switch rules and your function of X and Y; show how row, column and layer counts and spacing control both array size and module dimensions.
- **Visual communication - 30%:** the three sheets reveal repetition, façade patterns and the cube volume; captions and diagrams follow the Graphic Standards Manual.
- **Technical execution - 25%:** changing counts and spacing in each array rebuilds the copies and adapts each panel or cube to its new cell without manual remodeling or repositioning. The intended fit or gap is preserved, totals are correct, rules affect the intended copies, and saved inputs reproduce every output. Sheets 1-2 use the panel and Sheet 3 uses layered cubes.
- **Exploration - 15%:** four distinct façade patterns and purposeful tests of the function support the result you choose. Voids remain optional.

## Before You Submit

- [ ] Sheet 1 has exactly four diagrams showing the 3.1 module alone and in 1D, 2D and 3D arrays.
- [ ] Axes, counts, spacing and totals are labeled for the basic arrays.
- [ ] Changing row, column and layer counts and spacing in the reopened graph updates every array and the module or cube dimensions without manual edits.
- [ ] Sheet 2 has four different façade patterns of your module, each with its if/Switch rule and what it does.
- [ ] Sheet 3 shows the cube volume in isometric, with several layers and heights that change with X and Y.
- [ ] The height function and the chosen values are recorded; any optional void rule is described.
- [ ] Three sheets submitted together; the file reopens and reproduces everything shown.
- [ ] Library or self-made materials identified; textures packed.

## 3.3 - Attractor-driven panel façade

**Introduced:** Week 9 - Oct 20 / **Due:** Week 13 - Nov 17, 11:59 PM Eastern Time / **Weight:** 10%

**Preparation checks:** Week 10 - Oct 27: Bring a uniform façade and a first test in which one curve and one or two point attractors change one panel dimension.

### Description

Design a façade that responds to one curve attractor and one or two point attractors together. Combine their influence to change one dimension of your 3.1 panel. Show the whole composition in elevation and its three-dimensional result in isometric.

### Rules & Constraints

- Use your 3.1 panel and the 2D array from 3.2 to build a façade. Use one curve attractor with one or two point attractors. For each panel, measure the shortest distance to the curve and the distance to each point.
- Use the attractors to drive exactly one dimension of the panel, such as its opening (aperture), fold depth or corner offset, and rebuild the panel geometry from it. Changing only color, or only moving unchanged panels, does not count.
- Rescale (remap) the curve and point distances into influence values and combine them with a rule you state, such as an average or a weighted mix. The curve and the point attractor(s) must both contribute. Diagram how you combine them and limit (clamp) the result to the parameter range you tested in 3.1; keep the panel's other inputs fixed.
- Compare a uniform façade (no attractors) with the combined-attractor façade. Show one more version by moving one point or editing the curve, keeping everything else the same.
- Present one 17 x 11 inch composition with a façade elevation and an isometric of the same final façade. Show the curve, every point attractor and their combined influence - as an overlay or a separate diagram - with a legend.
- Submit a print-ready PDF and the editable graph. These are views of a 3D panel system, not a flat 2D line drawing. Pen plotting is not required.

### Parameters

- Façade row and column counts, panel spacing and base plane/origin. Keep the grid fixed so the comparison only shows changes in the panels.
- One attractor curve and one or two attractor points. Record the distance ranges, falloff, how you combine them and any weights; limit (clamp) the combined influence before it changes the panel.
- The one panel dimension you vary, its units and its minimum and maximum. Keep the other shape parameters fixed, and test that the panel is valid at both extremes.

### Required Studies

- **Baseline.** Show a small uniform façade with the selected panel parameter fixed. Label the constant value.
- **From distance to form.** Diagram the chain: distances to the curve and points to rescaled influences to combined to limited to the panel's range to panel dimension. Show the combined field and the curve and point locations, with a clear legend.
- **One change at a time.** Move one point or edit the curve once. Keep the grid, the other attractors, the mapping and the other panel inputs the same, so the effect is clear.
- **Final composition.** Give the elevation and matching isometric the most space. Label the curve, all point attractors, the panel dimension they drive and the final settings. Use your Graphic Standards Manual.

### Workflow

1. Bring in your own panel and make a small façade grid. Check the uniform version first.
2. Measure each panel's distance to the curve and to the point(s). Rescale and combine their influence, limit it to the chosen range, and generate the panels. Check that each attractor actually changes the result.
3. Check the nearest and farthest panels and several in between. Fix gaps, self-intersections or broken faces before you increase the number of panels.

4. Capture the two combined-attractor versions, and lay out the elevation, isometric and diagrams from the saved final settings.

## Deliverables

- **One 17 x 11 inch landscape PDF:** Lastname\_Firstname\_P3c.pdf. Include the final façade elevation and isometric, the curve and point locations, combined-field legend, mapping diagram, baseline and one changed-attractor configuration.
- **Editable file** in Lastname\_Firstname\_P3c/: save the panel, array, curve, point(s), combination and mapping controls; pack all files and materials it uses. Record the settings for the final views.
- Set up the composition to print normally at its stated size. No pen-plotter output or physical façade model is required.

## Evaluation

- **Computational understanding - 30%:** explain how the curve and point influences combine to change one panel dimension, and show what happens when you move a point or edit the curve.
- **Visual communication - 30%:** elevation, isometric, field legend and captions explain the façade composition and follow the Graphic Standards Manual.
- **Technical execution - 25%:** one curve and one or two point attractors all change the panels, which stay valid; the packed file and saved settings reproduce the final views.
- **Exploration - 15%:** the uniform and changed-attractor studies show clearly what each control does and support your final composition.

## Before You Submit

- [ ] Your own panel and array reused; one curve and one or two point attractors together change one panel dimension.
- [ ] The curve and point locations, distance ranges, combination rule, units and limits are documented.
- [ ] The uniform version and one other attractor setting are shown.
- [ ] The final elevation and isometric show the same façade; the attractor influence is easy to read.
- [ ] Print-ready composition, and a packed file that reproduces the final result.

**Shared standards:** Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages follow the instructor review draft. This handout uses US Letter; student sheets are 17 x 11 inches.

**Midterm grade:** The midterm grade accumulates the Graphic Standards Manual, CSG sheet and print, module (3.1) and arrays (3.2). These assignments account for 40% of the final course grade. Report the accumulated result as a percentage of those 40 possible course points. The midterm adds no separate assessment weight.